Power Supply

LiPo Charger

+5V
CHARGING1
LED/YELLOW/0603
R2
2.2k/R0402
R1
10k/R0402
C1
22uF/6.3V/20%/X5R/C0603
U1
VCC VBAT
CHRGb
BATTERY CHARGER
GND PROG
BL4054B-42TPRN(SOT23-5)
R4
Rprog
10k/R0402
I(Bat)
I(Bat) = 1000/Rprog = 100mA
Recommended LI-PO battery:
https://www.olimex.com/Products/Power/BATTERY-LIPO1400mAh/
GND

FET1
WPM2015-3/TR
D3
G1
S2
D1
VBAT
BAT1
3.7V LI-PO Battery Connector
DW02S
Voltage & Polarity!!!
PWR_FLAG
D2
1N5819S4/SOD123
C3
22uF/6.3V/20%/X5R/C0603

Battery Measurement

VBAT
R6
470k/R0402
Vref(ADC) = 1.1V!
R7
150k/R0402
C4
100nF/10V/10%/X7R/C0402
GPIO6\BAT_SENS

External Power Sense

+5V
R8
220k/R0402
R9
470k/R0402
GPIO5\PWR_SENS

U2
SY8089AAAC(SOT23-5)
Vout=0.6*(1+Ra/Rb)
L1
2.2uH/1.5A/DCR=72mR/20%/3.00x3.00x1.50mm/CIG32(WE3015T2R2M)
R10
Ra
220k/R0402
R11
Rb
49.9k/1%/R0402
C5
22uF/6.3V/20%/X5R/C0603
PWR_FLAG
+3.3V

USB to UART

USB-UART1
PWR_FLAG
+5V
D1
1N5819S4/SOD123
VBUS A4 B9 B4 A9
DN1 A7
DN2 B7
DP1 A6
DP2 B6
CC1 A5
CC2 B5
SBU1 A8
SBU2 B8
GND A1 B12 B1 A12
SHELL 0
R3 5.1k/R0603
R5 5.1k/R0603
USB-C-16P-F(A40-00119-A52-12)

Auto program			
DTR	RTS->EN	IO0	
1	1	1	1
0	0	1	1
1	0	0	1
0	1	1	0

U3
CH340X(MSOP-10)
VCC TXD
V3 RXD
GND RTS#
TNOW/DTR#
UD- CTS#
UD+
C2
100nF/10V/10%/X7R/C0402
R24
100R/R0402
GPIO44\U0RXD
R12
4.7k/R0402
GPIO43\U0TXD
UD-
UD+
R22
1k/R0402
R23
1k/R0402
R27
NA/R0402
T1
T2
BC817-40(SOT23)
GPIO0\BUT1
ESP_EN

pUEXT

GPIO17\U1TXD
GPIO18\U1RXD
GPIO47\I2C_SCL
GPIO48\I2C_SDA
GPIO13\FSPIQ(SPI3_MISO)
GPIO11\FSPID(SPI3_MOSI)
GPIO12\FSPICLK(SPI3_CLK)
GPIO10\FSPICS0(SPI3_CS0)
+3.3V
GND
R19 2.2k/R0402
R20 2.2k/R0402
R21 10k/R0402
pUEXT1
BM10B-SRSS-TB

Extensions

J1

EXT1
+3.3V at Iout(max)=600mA,
Power Out!
+3.3V
+3.3V
ESP_EN
GPIO4
GPIO5\PWR_SENS
GPIO6\BAT_SENS
GPIO7
GPIO15
GPIO16
GPIO17\U1TXD
GPIO18\U1RXD
GPIO8
GPIO3
GPIO46
GPIO9
GPIO10\FSPICS0(SPI3_CS0)
GPIO11\FSPID(SPI3_MOSI)
GPIO12\FSPICLK(SPI3_CLK)
GPIO13\FSPIQ(SPI3_MISO)
GPIO14
+5V Only!!!
Power In/Out
+5V
GND
HN1x22

J3

EXT2
GPIO43\U0TXD
GPIO44\U0RXD
GPIO1
GPIO2
GPIO42
GPIO41
GPIO40
GPIO39
GPIO38\LED1
*NC GPIO37
*NC GPIO36
*NC GPIO35
GPIO0\BUT1
GPIO45
GPIO48\I2C_SDA
GPIO47\I2C_SCL
GPIO21
GPIO20\USB_D+
GPIO19\USB_D-
GND
HN1x22

Extension's pinout is fully compatible to the Espressif's ESP32-S3-DevKitC-1.

ESP32-S3-WROOM MODULE

ESPRESSIF
U4
Espressif
ESP32-S3-WROOM-1x
*In module variants that have embedded OSPI PSRAM, i.e., that embed ESP32-S3R8, pins IO35, IO36 and IO37 are connected to the OSPI PSRAM and are not available for other uses.
Note: ESP32-S3 has four strapping pins: GPIO0, GPIO3, GPIO45 and GPIO46. GPIO0, GPIO46 and GPIO45 are connected to the chip's internal weak pull-up/pull-down during the chip reset. GPIO3 is floating by default.

3V3		IO0/RTC_GPIO0/GPIO0	27 GPIO0\BUT1
		IO1/RTC_GPIO1/GPIO1/TOUCH1/ADC1_CH0	39 GPIO1
		IO2/RTC_GPIO2/GPIO2/TOUCH2/ADC1_CH1	38 GPIO2
		IO3/RTC_GPIO3/GPIO3/TOUCH3/ADC1_CH2	15 GPIO3
		IO4/RTC_GPIO4/GPIO4/TOUCH4/ADC1_CH3	4 GPIO4
		IO5/RTC_GPIO5/GPIO5/TOUCH5/ADC1_CH4	5 GPIO5\PWR_SENS
GND		IO6/RTC_GPIO6/GPIO6/TOUCH6/ADC1_CH5	6 GPIO6\BAT_SENS
GND		IO7/RTC_GPIO7/GPIO7/TOUCH7/ADC1_CH6	7 GPIO7
EPAD(GND)		IO8/RTC_GPIO8/GPIO8/TOUCH8/ADC1_CH7/SUBSPICS1	12 GPIO8
		IO9/RTC_GPIO9/GPIO9/TOUCH9/ADC1_CH8/FSPIHD/SUBSPIHD	17 GPIO9
		IO10/RTC_GPIO10/GPIO10/TOUCH10/ADC1_CH9/FSPICS0/FSPIIO4/SUBSPICS0	18 GPIO10\FSPICS0(SPI3_CS0)
		IO11/RTC_GPIO11/GPIO11/TOUCH11/ADC2_CH0/FSPID/FSPIIO5/SUBSPID	19 GPIO11\FSPID(SPI3_MOSI)
		IO12/RTC_GPIO12/GPIO12/TOUCH12/ADC2_CH1/FSPICLK/FSPIIO6/SUBSPICLK	20 GPIO12\FSPICLK(SPI3_CLK)
		IO13/RTC_GPIO13/GPIO13/TOUCH13/ADC2_CH2/FSPIQ/FSPIIO7/SUBSPIQ	21 GPIO13\FSPIQ(SPI3_MISO)
		IO14/RTC_GPIO14/GPIO14/TOUCH14/ADC2_CH3/FSPIWP/FSPIDQS/SUBSPIWP	22 GPIO14
		IO15/RTC_GPIO15/GPIO15/U0RTS/ADC2_CH4/XTAL_32K_P	8 GPIO15
		IO16/RTC_GPIO16/GPIO16/U0CTS/ADC2_CH5/XTAL_32K_N	9 GPIO16
EN		IO17/RTC_GPIO17/GPIO17/U1TXD/ADC2_CH6	10 GPIO17\U1TXD
		IO18/RTC_GPIO18/GPIO18/U1RXD/ADC2_CH7/CLK_OUT3	11 GPIO18\U1RXD
		IO19/RTC_GPIO19/GPIO19/U1RTS/ADC2_CH8/CLK_OUT2/USB_D-	13 GPIO19\USB_D-
		IO20/RTC_GPIO20/GPIO20/U1CTS/ADC2_CH9/CLK_OUT1/USB_D+	14 GPIO20\USB_D+
		IO21/RTC_GPIO21/GPIO21	23 GPIO21
		IO35*/SPIIO6/GPIO35/FSPID/SUBSPID	28 *NC GPIO35
		IO36*/SPIIO7/GPIO36/FSPICLK/SUBSPICLK	29 *NC GPIO36
		IO37*/SPIDQS/GPIO37/FSPIQ/SUBSPIQ	30 *NC GPIO37
		IO38/GPIO38/FSPIWP/SUBSPIWP	31 GPIO38\LED1
		IO39/MTCK/GPIO39/CLK_OUT3/SUBSPICS1	32 GPIO39
		IO40/MTDO/GPIO40/CLK_OUT2	33 GPIO40
		IO41/MTDI/GPIO41/CLK_OUT1	34 GPIO41
		IO42/MTMS/GPIO42	35 GPIO42
		TXD0/U0TXD/GPIO43/CLK_OUT1	37 GPIO43\U0TXD
		RXD0/U0RXD/GPIO44/CLK_OUT2	36 GPIO44\U0RXD
		IO45/GPIO45	26 GPIO45
		IO46/GPIO46	16 GPIO46
		IO47/SPICLK_P_DIFF/GPIO47/SUBSPICLK_P_DIFF	24 GPIO47\I2C_SCL
		IO48/SPICLK_N_DIFF/GPIO48/SUBSPICLK_N_DIFF	25 GPIO48\I2C_SDA

ESP32-S3-WROOM-1-N8R8

+3.3V
C6 22uF/6.3V/20%/X5R/C0603
C7 100nF/50V/10%/X7R/C0402
GND
+3.3V
R14 10k/R0402
ESP_EN
C8 1uF/10V/10%/X5R/C0603
PWR_FLAG
GND
R18 10k/R0402
+3.3V
GPIO3 when
1: JTAG signal from USB Serial/JTAG controller
0: JTAG signal from on-chip JTAG pins

USER LED

+3.3V
R13 2.2k/R0402
LED1 LED/Green/0603
GPIO38\LED1

Fiducials

FID1 Fiducial
FID2 Fiducial
FID3 Fiducial
FID4 Fiducial

Reset Button

ESP_EN
RST1 YTS-A016-X
GND

USER Button

GPIO0\BUT1
R15 100R/R0402
BUT1 YTS-A016-X
GND

USB-OTG

Full-speed USB 1.1 On-The-Go (OTG) interface or USB Serial/JTAG controller.
The ESP32-S3 contains an USB Serial/JTAG Controller. This unit can be used to program the SoC's flash, read program output, as well as attach a debugger to the running program.
The ESP32-S3 has an Internal PHY, which is shared between the USB-OTG and the USB Serial/JTAG hardware. Either one of these can use the internal PHY.

+5V
PWR_FLAG
D3 1N5819S4/SOD123
USB-OTG1
VBUS A4 B9 B4 A9
GPIO19\USB_D- A7 B7 DN1 DN2
GPIO20\USB_D+ A6 B6 DP1 DP2
CC1 A5
CC2 B5
SBU1 A8
SBU2 B8
GND A1 B12 B1 A12
SHELL 0
R16 5.1k/R0603
R17 5.1k/R0603
USB-C-16P-F(A40-00119-A52-12)
GND

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open source hardware

Sheet: /
File: ESP32-S3-DevKit-LiPo_Rev_B.sch
Title: ESP32-S3-DevKit-LiPo
Size: A3 | Date: 2023-05-16 | Rev: B
KiCad E.D.A. kicad 5.1.0-rc2-unknown-036be7d80ubuntu16.04.1 | Id: 1/1